

MA 107: Mathematics-II-QUIZ-4 - UG

Name :

Roll No :

Sec :

Teacher Name :

Instructions: Attempt all questions. Only the response marked on this sheet will be evaluated. Time: one hour

1. Consider a random variable  $X$  whose pmf  $p(x)$  is positive at  $x = -1, 0, 1$  and zero elsewhere. If  $p(0) = \frac{1}{4}$ , then  $E(X^2)$  equals.....  $3/4$

2. Consider a random variable  $X$  whose pdf is given by  $f(x) = \frac{1}{2}e^{-|x|}$ ,  $-\infty < x < \infty$ . Assuming the MGF of  $X$  exists, the Moment Generating Function (MGF) is given by

- (a)  $1/(1-t^2)$  (b)  $1/(t^2-1)$  (c)  $t/(1-t)$  (d) None of these

3. Let  $X$  be a continuous random variable with probability density function, given by  $f(x) = \begin{cases} \theta x + \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$ ,  $\theta$  is constant. The value of  $\theta$  for which variance of  $X$  is maximum is

- (a) 1 (b) 0 (c)  $\frac{1}{4}$  (d) -1

4. A normal random variable  $X$  has the following probability density function

$$f(x) = \frac{1}{\sqrt{8\pi}} e^{-\left(\frac{x-1}{8}\right)^2}, \quad -\infty < x < \infty.$$

Then  $\left(\int_1^\infty f(x)dx\right)$  is .....  $1/2$

5. Minimum number of throws of an unbiased die necessary to ensure that the probability of getting a six at least once is greater than or equal to  $\frac{1}{2}$  is.....  $4$

6. Let  $X$  be a non-negative random variable with  $E(X^2) = 3$  and  $E(X) = 1$ , Then  $\sum_{i=1}^{\infty} i P(X \geq i)$  is

- (a) 1 (b) 3 (c) 2 (d) 4

7. Value of  $\mathbb{P}(|X| \leq 1/3) + \mathbb{P}\left(|X| = \frac{1}{2}\right)$  if cdf of  $X$  is given as

$$F(x) = \begin{cases} 0 & \text{if } x < -1 \\ \frac{x+1}{2} & \text{if } -1 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

Ans.....  $1/3$

8.  $X \sim \text{Bin}(n, p)$  with  $n > 15, p > 0$ . Then  $n - X \sim \dots \text{B}(n, 1)$

9. Let  $X$  be a random variable which is symmetric about 0. Let  $F$  be the cumulative distribution function of  $X$ . Which of the following statements is NECESSARILY/ALWAYS true  $\forall x \in \mathbb{R}$ ?

(a)  $F(x) + F(-x) = 1$

(b)  $F(x) - F(-x) = 1$

(c)  $F(x) + F(-x) = 1 + \mathbb{P}(X = x)$

(d)  $F(x) + F(-x) = 1 - \mathbb{P}(X = x)$

10.  $Y = \log_2 X$  follows  $N(\mu, \sigma^2)$  and  $M_Y(t) = e^{3t+3t^2}$ .  $Pr(X < 8)$

(a)  $\frac{1}{3}$

(b)  $\frac{1}{2}$

(c)  $\frac{3}{4}$

(d) None of the above